CS241 - Questions for Lab Assignment 6

1 INSTRUCTIONS

(1) This lab **is graded**.
(2) Use the student database that was created in Lab2.
(3) Instructions for downloading JDBC driver for MySQL:
  (a) Download JDBC connector from https://dev.mysql.com/downloads/connector/j/
  For Linux: Install using sudo apt install <.deb> file
  (b) Your .jar will be in /usr/share/java
  Set your CLASSPATH using the following command (for bash):
  `export CLASSPATH=$CLASSPATH:/usr/share/java/mysql-connector-java-8.0.15.jar`
  (c) To start a connection, use the following code:

```
String url = "jdbc:mysql://localhost:3306/<your database name>";
String username = "<your user name>";
String password = "<your password>";
Connection conn= DriverManager.getConnection(url, username, password);
```

(4) Do not use any frameworks for this assignment.
(5) 50

2 QUESTION 1

Consider the following relations:

Student(snum: integer, sname: string, major: string, level: string, age: integer)
Class(name: string, meets_at: time, room: string, fid: integer)
Enrolled(snum: integer, cname: string)
Faculty (fid: integer, fname: string, deptid: integer)

The meaning of these relations is straightforward; for example, Enrolled has one record per student-class pair such that the student is enrolled in the class.

(1) Write a Java application that lists all the tables and asks the user to select a table to print. Upon selection, the application must print the contents of the selected table in tabular form. [5 Marks]
(2) Write a Java application that runs the queries given below. Your application must handle error cases. If a query returns an error, the error must be displayed to the user. (Alternatively, you may write a separate application for each query.) [3*5=15 Marks]
  (a) Ask the user to specify the name of a room. Print the names of all classes that meet in that room.
  (b) Ask the user to specify the name of a course. Print the rooms and the times at which classes for that course will be conducted.
  (c) Ask the user to specify the name of a faculty member. Print the names of the courses that that faculty member teaches.

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